

Test Verification Summary

The table below shows pass (✓), fail (✗), skip (○), or not-applicable (—) for each of the 11 test categories across all cases. Results are produced by `test_milp.py` calling `baselines/milp.py`.

Case	Budget alloc.	Progress	EVM signals	Term. counter	MS cert.	Payment id.	Cash bal.	Z^*	Outcome	P monotone	Advance
<i>Group 1: Progress and Milestone Certification</i>											
SP-1	—	—	—	—	—	—	—	—	—	—	—
SP-3	—	—	—	—	—	—	—	—	—	—	—
SP-J	—	—	—	—	—	—	—	—	—	—	—
<i>Group 2: Discounting and Timing Incentives</i>											
SP-4	—	—	—	—	—	—	—	—	—	—	—
MP-K	—	—	—	—	—	—	—	—	—	—	—
MP-Kp	—	—	—	—	—	—	—	—	—	—	—
SP-L	—	—	—	—	—	—	—	—	—	—	—
SP-Lp	—	—	—	—	—	—	—	—	—	—	—
SP-M	—	—	—	—	—	—	—	—	—	—	—
SP-Mp	—	—	—	—	—	—	—	—	—	—	—
<i>Group 3: Advance Payments and Recovery</i>											
SP-2	—	—	—	—	—	—	—	—	—	—	—
SP-2a	—	—	—	—	—	—	—	—	—	—	—
SP-2b	—	—	—	—	—	—	—	—	—	—	—
SP-2c	—	—	—	—	—	—	—	—	—	—	—
SP-2d	—	—	—	—	—	—	—	—	—	—	—
SP-A	—	—	—	—	—	—	—	—	—	—	—
SP-B	—	—	—	—	—	—	—	—	—	—	—
<i>Group 4: Retention</i>											
SP-C	—	—	—	—	—	—	—	—	—	—	—
SP-D	—	—	—	—	—	—	—	—	—	—	—
<i>Group 5: Counter-Trajectory Cases (Active)</i>											
SP-E	—	—	—	—	—	—	—	○	—	—	—
SP-F	—	—	—	—	—	—	—	○	—	—	—
SP-G	—	—	—	—	—	—	—	○	—	—	—
<i>Group 6: Forced and Starvation Termination</i>											
SP-5	—	—	—	—	—	—	—	—	—	—	—
SP-H	—	—	—	—	—	—	—	—	—	—	—
SP-I	—	—	—	—	—	—	—	○	—	—	—
<i>Group 7: Multi-Project Mixed Outcomes</i>											
MP-3	—	—	—	—	—	—	—	—	—	—	—
MP-A	—	—	—	—	—	—	—	—	—	—	—
MP-B	—	—	—	—	—	—	—	—	—	—	—
<i>Group 8: Multi-Project Portfolio Optimisation</i>											
MP-2	—	—	—	—	—	—	—	—	—	—	—

continued ...

Case	<i>Budget alloc.</i>	<i>Progress</i>	<i>EVM signals</i>	<i>Term. counter</i>	<i>MS cert.</i>	<i>Payment id.</i>	<i>Cash bal.</i>	<i>Z*</i>	<i>Outcome</i>	<i>P monotone</i>	<i>Advance</i>
MP-4	—	—	—	—	—	—	—	—	—	—	—
MP-C	—	—	—	—	—	—	—	—	—	—	—
MP-D	—	—	—	—	—	—	—	—	—	—	—
<i>Group 9: Staggered-Start Multi-Project</i>											
MP-1	—	—	—	—	—	—	—	—	—	—	—
MP-E	—	—	—	—	—	—	—	—	—	—	—

Figure Legends

All figures share the following visual language.

Panel	Meaning
EVM (left)	 BCWS  BCWP  ACWP  θ_j thresholds
	Red dashed vertical = termination period.
Profile (centre)	 $x_{i,t}$ spend (down)  Inflows: R^{net} , A_i , R^{ret} (up)
SPI/CPI (right)	 = SPI (green),  = CPI (orange), red dashed = reference 1.
Portfolio left	Cyan step-line = B_t ; grey dashed = B_1 reference.
Portfolio centre	Green bars = inflow, red bars = outflow, blue line with $\circ$ = net per period.
Portfolio right	Blue line with $\bullet$ = cumulative discounted NCF.

X-axes scale to each case's horizon H . Middle project panel uses symmetric $\pm \text{BAC}_i$ range. Portfolio panels use data-driven ranges.

Conventions

- $\gamma = 0.95$ throughout.
- Advance A_i arrives at $t = s_i - 1$ (period 0 when $s_i = 1$).
- Payment identity: $A_i + \sum_j R_{i,j}^{\text{net}} + R_i^{\text{ret}} = CP_i$ checked for completed projects only.
- Retention released as a lump at the period the final milestone certifies.
- Advance recovery deductions: $\delta^{\text{rec}} \phi_j CP_i$ (pre-retention base, FIDIC convention).

Cases

Group 1: Progress and Milestone Certification

SP-1 (Group G1)

n, H	1, 4
BAC, CP	100, 120
$s_i, f_i, D^{\text{plan}}$	1, 4, 4
$\bar{\eta}$	1
M, θ	2, (0.5, 1)
$\phi, e_{i,j}$	(0.5, 0.5), (1, 3)
$\mu, \Omega, \tau^{\text{tol}}$	0.3, 1, 2
B_1, γ	300, 0.95
Z^*	19.025
Outcome	completed

t	x	P	BCWS	BCWP	ACWP	SPI	CPI	τ	MS	R_{net}
1	50	0.5	0.25	0.5	0.5	2	1	2	1	60
2	0	0.5	0.5	0.5	0.5	1	1	2	--	0
3	50	1	0.75	1	1	1.333	1	2	2	60
4	0	1	1	1	1	1	1	2	--	0

Portfolio strip.

t	B_t	ΣIn	ΣOut	ΣNet	$\gamma^{t-1} \text{Net}$	Z_t^{cum}
1	310	60	50	10	10	10
2	310	0	0	0	0	10
3	320	60	50	10	9.025	19.025
4	320	0	0	0	0	19.025

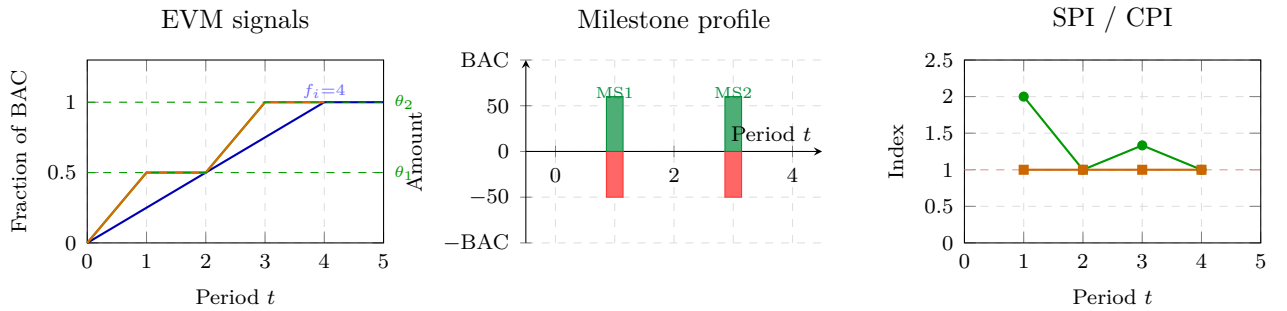


Figure 1: SP-1 project view (single-project). Outcome: **completed**. $Z^* = 19.025$.

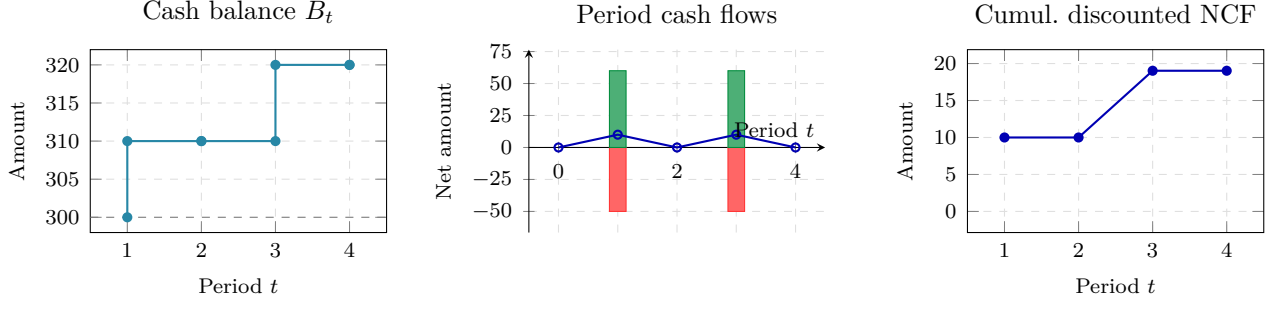


Figure 2: SP-1 portfolio view. $Z^* = 19.025$.

SP-3 (Group G1)

n, H	1, 6
BAC, CP	60, 90
$s_i, f_i, D^{\text{plan}}$	1, 6, 6
$\bar{\eta}$	1
M, θ	3, (0.333, 0.667, 1)
$\phi, e_{i,j}$	(0.333, 0.333, 0.333), (1, 3, 5)
$\mu, \Omega, \tau^{\text{tol}}$	0.5, 2, 2
B_1, γ	300, 0.95
Z^*	27.17
Outcome	completed

t	x	P	BCWS	BCWP	ACWP	SPI	CPI	τ	MS	R_{net}
1	20	0.333	0.167	0.333	0.333	2	1	2	1	30
2	0	0.333	0.333	0.333	0.333	1	1	2	--	0
3	20	0.667	0.5	0.667	0.667	1.333	1	2	2	30
4	0	0.667	0.667	0.667	0.667	1	1	2	--	0
5	20	1	0.833	1	1	1.2	1	2	3	30
6	0	1	1	1	1	1	1	2	--	0

Portfolio strip.

t	B_t	ΣIn	ΣOut	ΣNet	$\gamma^{t-1} \text{Net}$	Z_t^{cum}
1	310	30	20	10	10	10
2	310	0	0	0	0	10
3	320	30	20	10	9.025	19.025
4	320	0	0	0	0	19.025
5	330	30	20	10	8.145	27.17
6	330	0	0	0	0	27.17

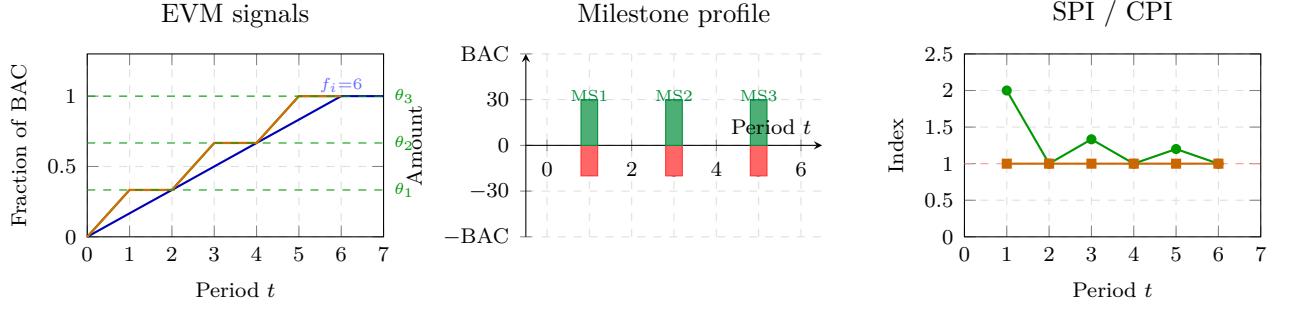


Figure 3: SP-3 project view (single-project). Outcome: **completed**. $Z^* = 27.17$.

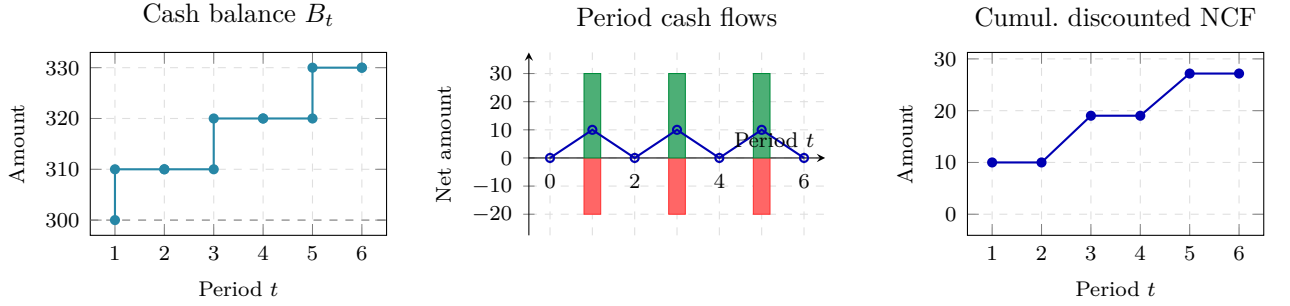


Figure 4: SP-3 portfolio view. $Z^* = 27.17$.

SP-J (Group G1)

n, H	1, 4
BAC, CP	100, 100
$s_i, f_i, D^{\text{plan}}$	1, 4, 4
$\bar{\eta}$	1
M, θ	2, (0.5, 1)
$\phi, e_{i,j}$	(0.5, 0.5), (1, 3)
$\mu, \Omega, \tau^{\text{tol}}$	0.5, 2, 2
B_1, γ	300, 0.95
Z^*	0
Outcome	completed

t	x	P	BCWS	BCWP	ACWP	SPI	CPI	τ	MS	R_{net}
1	50	0.5	0.25	0.5	0.5	2	1	2	1	50
2	0	0.5	0.5	0.5	0.5	1	1	2	--	0
3	50	1	0.75	1	1	1.333	1	2	2	50
4	0	1	1	1	1	1	1	2	--	0

Portfolio strip.

t	B_t	ΣIn	ΣOut	ΣNet	$\gamma^{t-1} \text{Net}$	Z_t^{cum}
1	300	50	50	0	0	0
2	300	0	0	0	0	0
3	300	50	50	0	0	0
4	300	0	0	0	0	0

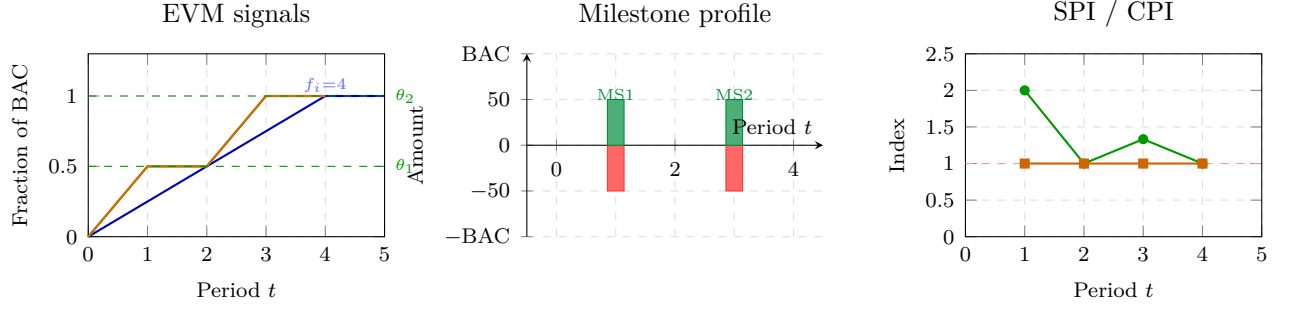


Figure 5: SP-J project view (single-project). Outcome: completed. $Z^* = 0$.

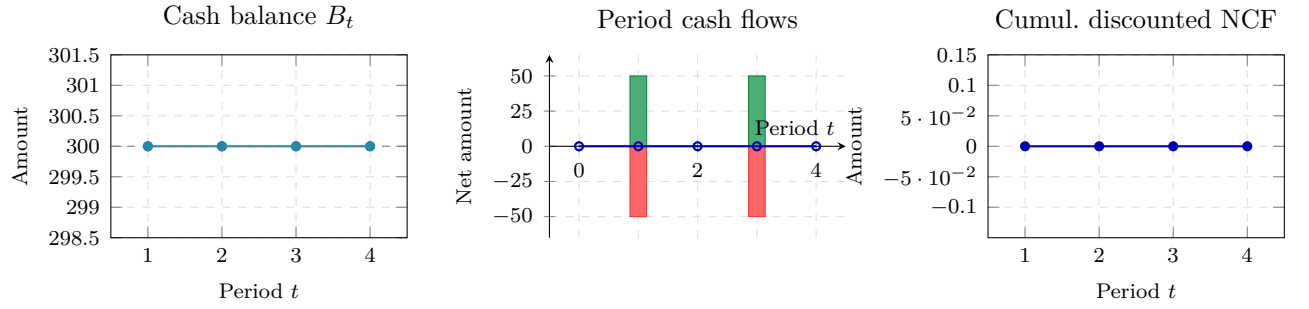


Figure 6: SP-J portfolio view. $Z^* = 0$.

Group 2: Discounting and Timing Incentives

SP-4 (Group G2)

n, H	1, 8
BAC, CP	100, 140
$s_i, f_i, D^{\text{plan}}$	1, 8, 8
$\bar{\eta}$	1
M, θ	2, (0.5, 1)
$\phi, e_{i,j}$	(0.5, 0.5), (1, 5)
$\mu, \Omega, \tau^{\text{tol}}$	0.3, 1, 2
B_1, γ	300, 0.95
Z^*	36.29
Outcome	completed

t	x	P	BCWS	BCWP	ACWP	SPI	CPI	τ	MS	R_{net}
1	50	0.5	0.125	0.5	0.5	4	1	2	1	70
2	0	0.5	0.25	0.5	0.5	2	1	2	--	0
3	0	0.5	0.375	0.5	0.5	1.333	1	2	--	0
4	0	0.5	0.5	0.5	0.5	1	1	2	--	0
5	50	1	0.625	1	1	1.6	1	2	2	70
6	0	1	0.75	1	1	1.333	1	2	--	0
7	0	1	0.875	1	1	1.143	1	2	--	0
8	0	1	1	1	1	1	1	2	--	0

Portfolio strip.

t	B_t	ΣIn	ΣOut	ΣNet	$\gamma^{t-1} \text{Net}$	Z_t^{cum}
1	320	70	50	20	20	20
2	320	0	0	0	0	20
3	320	0	0	0	0	20
4	320	0	0	0	0	20
5	340	70	50	20	16.29	36.29
6	340	0	0	0	0	36.29
7	340	0	0	0	0	36.29
8	340	0	0	0	0	36.29

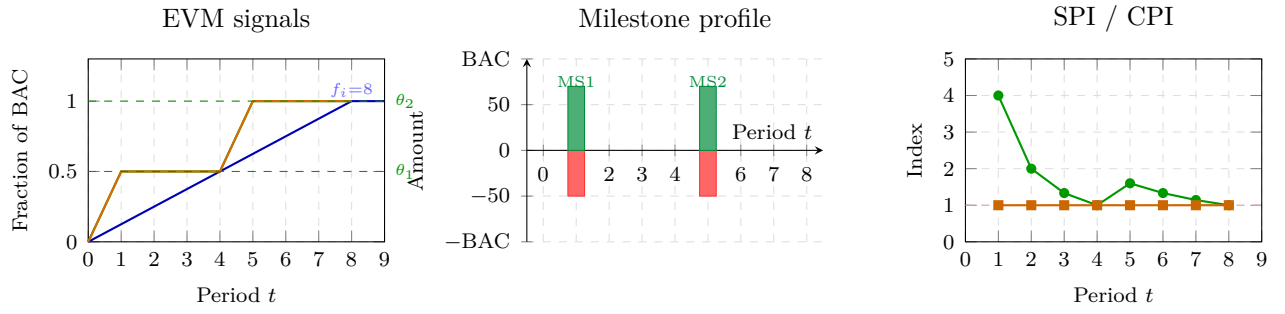


Figure 7: SP-4 project view (single-project). Outcome: **completed**. $Z^* = 36.29$.

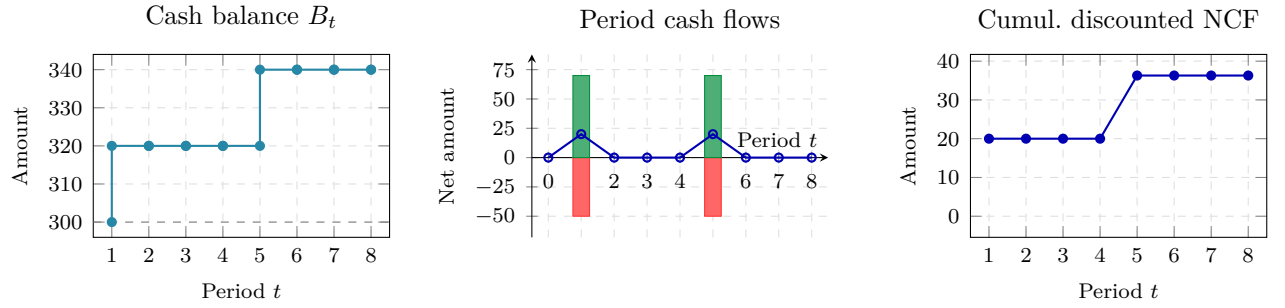


Figure 8: SP-4 portfolio view. $Z^* = 36.29$.

MP-K (Group G2)

n, H	2, 6
BAC_i	(60, 60)
CP_i	(90, 90)
B_1, γ	300, 0.95
Z^*	55.87
Outcome	both_completed

Project 1 timeline.

t	x	P	BCWS	BCWP	ACWP	SPI	CPI	τ	MS	R_{net}
1	30	0.5	0.167	0.5	0.5	3	1	2	1	63
2	0	0.5	0.333	0.5	0.5	1.5	1	2	--	0
3	0	0.5	0.5	0.5	0.5	1	1	2	--	0
4	30	1	0.667	1	1	1.5	1	2	2	27
5	0	1	0.833	1	1	1.2	1	2	--	0
6	0	1	1	1	1	1	1	2	--	0

Project 2 timeline.

t	x	P	BCWS	BCWP	ACWP	SPI	CPI	τ	MS	R_{net}
1	0	0	0.167	0	0	0	1	2	--	0
2	30	0.5	0.333	0.5	0.5	1.5	1	2	1	27
3	0	0.5	0.5	0.5	0.5	1	1	2	--	0
4	30	1	0.667	1	1	1.5	1	2	2	63
5	0	1	0.833	1	1	1.2	1	2	--	0
6	0	1	1	1	1	1	1	2	--	0

Portfolio strip.

t	B_t	ΣIn	ΣOut	ΣNet	$\gamma^{t-1} \text{Net}$	Z_t^{cum}
1	333	63	30	33	33	33
2	330	27	30	-3	-2.85	30.15
3	330	0	0	0	0	30.15
4	360	90	60	30	25.721	55.871
5	360	0	0	0	0	55.871
6	360	0	0	0	0	55.871

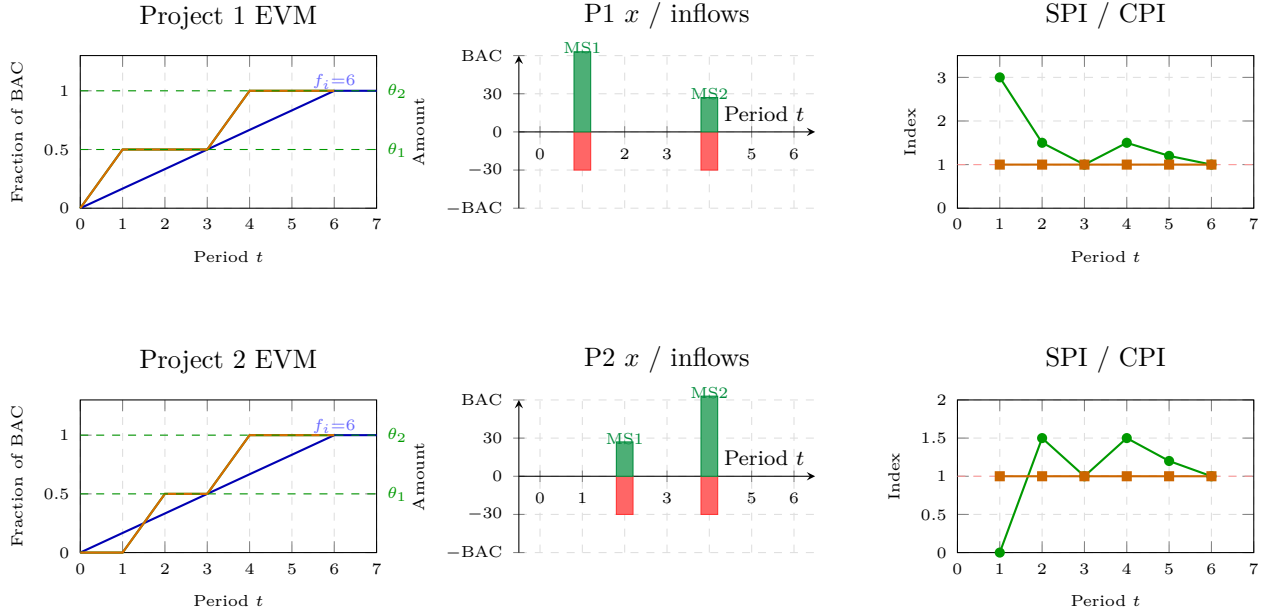


Figure 9: MP-K project view (2-project). Outcome: $\text{both}_{\text{completed}}.Z^* = 55.87$.